

1-4-1: Comparing TCP_IP to the OSI Model

At the end of this episode, I will be able to:

1. Compare and contrast the TCP/IP and OSI Models, including functionality and layer structure.

Learner Objective: *Compare and contrast the TCP/IP and OSI Models, including functionality and layer structure.

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Description: In this episode, the learner will examine the similarities, differences, and functionalities of the TCP/IP and OSI Model layers.

- Comparison of TCP/IP and OSI Models
 - o Structural Differences
 - ♣ TCP/IP model consists of four layers: Link, Internet, Transport, Application.
 - ♣ OSI model has seven layers: Physical, Data Link, Network, Transport, Session, Presentation, Application.
 - o Focus and Approach
 - ♣ TCP/IP is more practical, used in real-world network implementations and protocol development.
 - ♣ OSI is theoretical, serving as a detailed reference model for understanding network functions.
 - o Usage and Application
 - ♣ TCP/IP is widely adopted in actual network communication systems.
 - ♣ OSI model is used more as an educational tool to explain and understand how different network layers interact.
 - Comparison of the layers
 - o Application Layer
 - ♣ TCP/IP - Combines OSI's Application, Presentation, and Session layers, handling high-level protocols, representation, and session management.
 - ♣ OSI - Separate Application, Presentation, and Session layers, with more distinct functions for each layer.
 - o Transport Layer
 - ♣ TCP/IP Transport Layer deals with end-to-end communication (TCP, UDP).
 - ♣ OSI Transport Layer ensures reliable data transfer and correct data sequencing.
 - o Internet/Network Layer
 - ♣ TCP/IP Internet Layer manages logical addressing and routing (IP).
 - ♣ OSI Network Layer handles routing, addressing, and data packet forwarding.
 - o Link/Network Interface Layer
 - ♣ TCP/IP Link Layer is responsible for physical network hardware and media access control.
 - ♣ OSI Data Link Layer provides node-to-node data transfer and error correction.
 - o OSI-Specific Layers
 - ♣ OSI Session Layer (absent in TCP/IP) manages sessions and dialogues between applications.
 - ♣ OSI Presentation Layer (absent in TCP/IP) translates data for the Application Layer.
 - ♣ OSI Physical Layer corresponds to the Network Interface Layer in TCP/IP, dealing with physical aspects of network communication.
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- Additional Resources:

° If applicable